The documentation and process conversion measures necessary to comply with this revision shall be completed by .31 November 1998

INCH-POUND

MIL-PRF-19500/291G
31 August 1998
SUPERSEDING
MIL-PRF-19500/291F
4 November 1997

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, SWITCHING,
TYPES 2N2906A, 2N2906AL, 2N2907A, 2N2907AL, 2N2906AUA
2N2907AUA, 2N2906AUB, AND 2N2907AUB
JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This Specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for PNP, silicon, switching transistors. Four levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500 and two levels of product assurance are provided for each unencapsulated device type.

1.2 Physical dimensions. See figure 1 (similar to a T0-18), figures 3 and 4 (surface mount case outlines UA and UB), and figures 5 and 6 (JANHC and JANKC).

1.3 Maximum ratings.

Types	P_T 1/ T_A = +25°C	P_T 2/,3/ T_C = +25°C	I_C	V_{CBO}	V_{CEO}	V_{EBO}	T_J and T_{STG}
	W	W	mA dc	V dc	V dc	V dc	°C
2N2906A, L, UA, UB	0.4 1/	1.8 2/ 3/	600	60	60	5	-65 to +200
2N2907A, L, UA, UB	0.4 1/	1.8 2/ 3/	600	60	60	5	-65 to +200

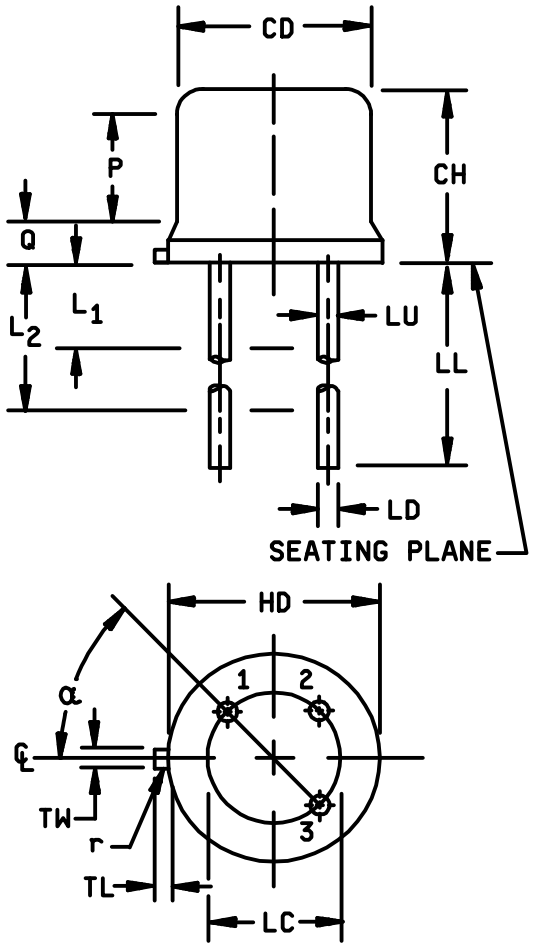
1/ Derate linearly 2.28 mW/°C for T_A > +25°C.
2/ Derate linearly 10.3 mW/°C for T_C > +25°C (Nonsurface mount devices).
3/ For UA and UB surface mount case outlines, P_T = 1.16 W; derate linearly 6.6 mW/°C for T_C > +25°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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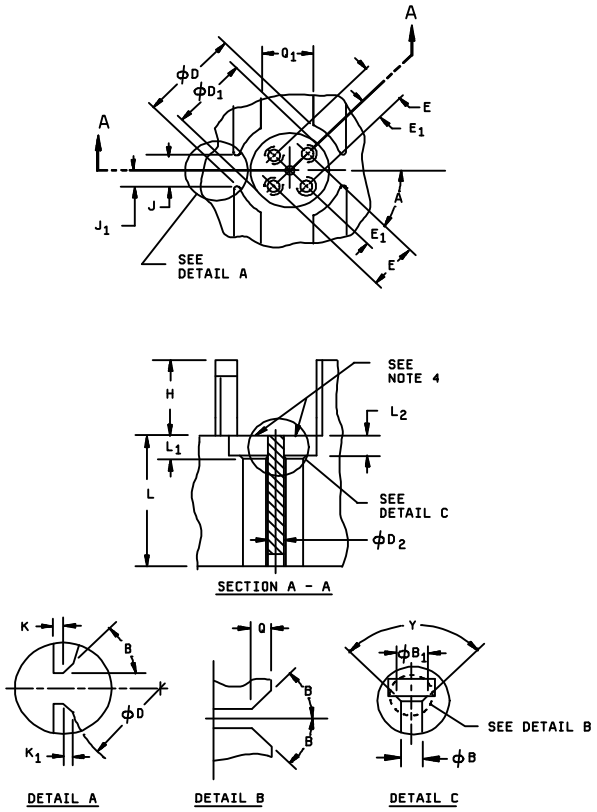
Symbol	Inches		Millimeters		Notes
	Min	Max	Min	Max	
CD	.178	.195	4.52	4.95	
CH	.170	.210	.8255	.8509	
HD	.209	.230	5.31	5.84	
LC	100 TP		2.54 TP		6
LD	.016	.021	0.41	0.53	7,8
LL	.500	.750	12.70	19.05	7,8,13
LU	.016	.019	0.41	0.48	7,8
L_1	---	.050	---	1.27	7,8
L_2	.250	---	6.35	---	7,8
Q	---	.030	----	0.76	5
TL	.028	.048	0.71	1.22	3,4
TW	.036	.046	0.91	1.17	3
r	---	.010	---	0.25	10
"	45° TP		45° TP		6
P	.100		2.54		



NOTES:
1. Dimension are in inches.
2. Metric equivalents are given for general information only.
3. Beyond r (radius) maximum, TL shall be held for a minimum length of .011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by the gauge and gauging procedure shown in figure 2.
7. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
8. All three leads.
9. The collector shall be internally connected to the case.
10. Dimension r (radius) applies to both inside corners of tab.
11. In accordance with ANSI Y14.5M, diameters are equivalent to Nx symbology.
12. Lead 1 = emitter, lead 2 = base, lead 3 = collector.
13. For L suffix devices, dimension LL = 1.5 inches (38.10 mm) min. and 1.75 inches (44.45 mm) max.

FIGURE 1. Physical dimensions (similar to TO-18).

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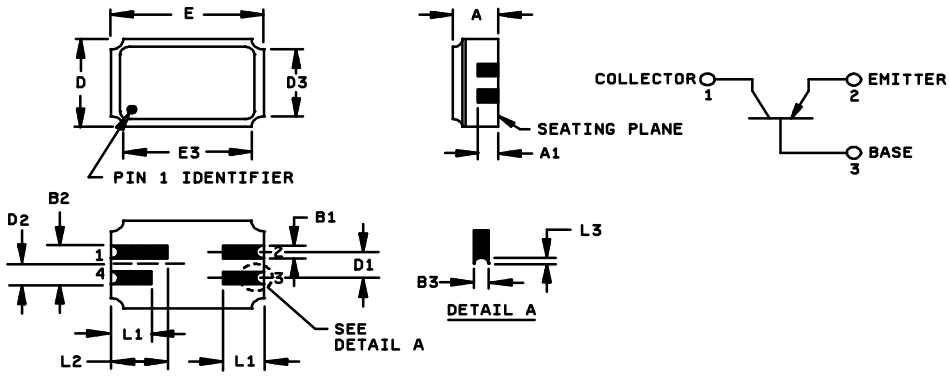
Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
NB	.8255	.8509	.0325	.0335	5
NB_1	1.09 Nom		.040 Nom		5
ND	5.867	5.880	.2310	.2315	
ND_1	4.04	4.09	.159	.161	
ND_2	1.02 Nom		.040 Nom		6
E	2.527	2.553	.0995	.1005	
E_1	1.257	1.283	.0495	.0505	
H	3.68	3.94	.145	.155	
J	1.194	1.207	.0470	.0475	
J_1	.597	.622	.0235	.0245	

Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
K	.229	.279	.009	.011	
K_1	.127 Nom		.005 Nom		
L	9.45	9.60	.372	.378	
L_1	1.37	1.40	.054	.055	
L_2	1.09 Nom		.043 Nom		
Q	1.02 Nom		.040 Nom		
Q_1	3.12	3.23	.123	.127	
"	---	---	44.90E	45.10E	
B	---		45E Nom		
Y	---		90E Nom		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The following gauging procedures shall be used: The device being measured shall be inserted until its seating plane is .125 inch (3.18 mm) ±.010 inch (0.254 mm) from the seating surface of the gauge. A force of 8 ±.5 ounces shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gauge. The use of a pin straightener prior to insertion in the gauge is permissible. A spacer may be used to obtain the .125 inch (3.18 mm) distance from the gauge seat prior to force application.
4. These surfaces to be parallel and in same plane within ± .001 inch (0.025 mm).
5. Four holes.
6. Pressed in.

FIGURE 2. Gauge for lead and tab location.

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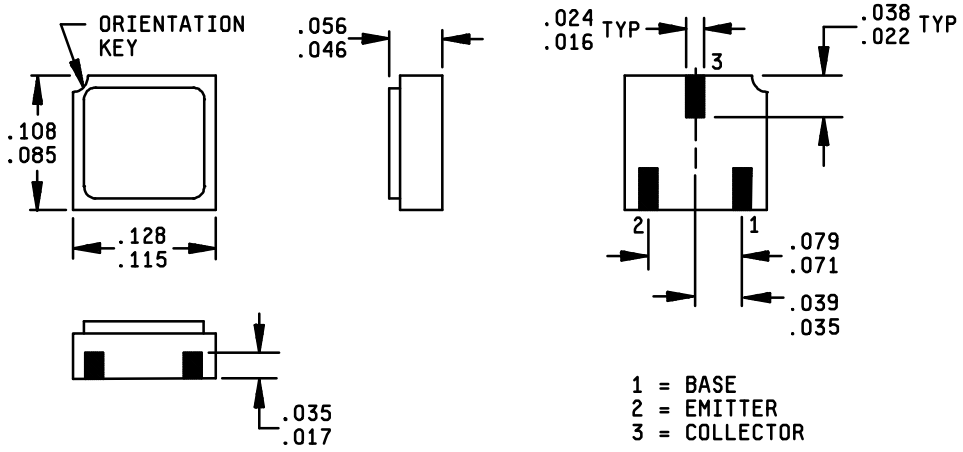
Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.061	.075	1.55	1.90	3
A_1	.029	.041	0.74	1.04	
B_1	.022	.028	0.56	0.71	
B_2	.075 REF		1.91 REF		
B_3	.006	.022	0.15	0.56	5
D	.145	.155	3.68	3.93	
D_1	.045	.055	1.14	1.39	

Ltr	Dimensions				Notes
	Inches		Millimeter		
	Min	Max	Min	Max	
D_2	.0375 BSC		.952 BSC		
D_3	---	.155	---	3.93	
E	.215	.225	5.46	5.71	
E_3	--	.225	--	5.71	
L_1	.032	.048	0.81	1.22	
L_2	.072	.088	1.83	2.23	
L_3	.003	.007	0.08	0.18	5

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimension "A" controls the overall package thickness. When a window lid is used, dimension "A" must increase by a minimum of .010 inch (0.254 mm) and a maximum of .040 inch (1.020 mm).
4. The corner shape (square, notch, radius, etc.) may vary at the manufacturer's option, from that shown on the drawing.
5. Dimensions "B3" minimum and "L3" minimum and the appropriately castellation length define an unobstructed three-dimensional space traversing all of the ceramic layers in which a castellation was designed. (Castellations are required on bottom two layers, optional on top ceramic layer.) Dimension "B3" maximum and "L3" maximum define the maximum width and depth of the castellation at any point on its surface. Measurement of these dimensions may be made prior to solder dipping.

FIGURE 3. Physical dimensions, surface mount (UA version).

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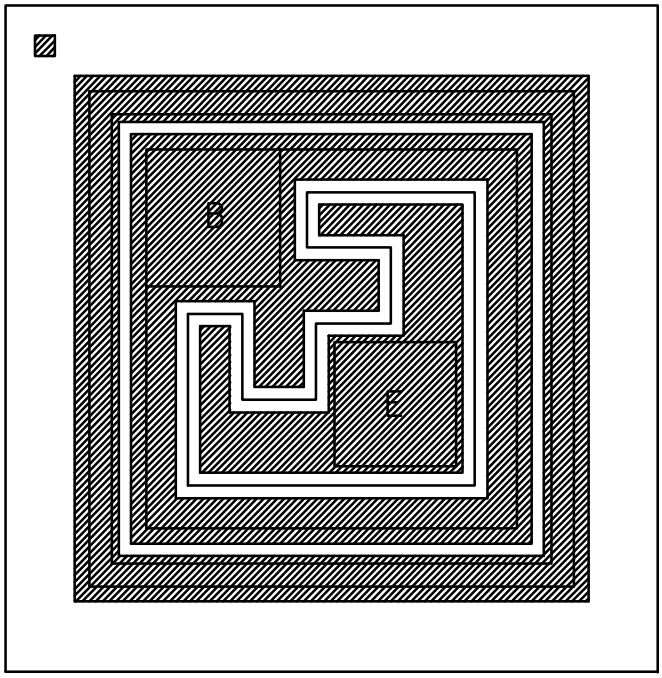


Inches	mm	Inches	mm
.016	0.41	.056	1.42
.017	0.43	.071	1.81
.022	0.56	.079	2.01
.024	0.61	.085	2.41
.035	0.89	.108	2.74
.038	0.96	.115	2.82
.039	0.99	.128	3.25
.046	0.97		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 4. Physical dimensions, surface mount (UB version).

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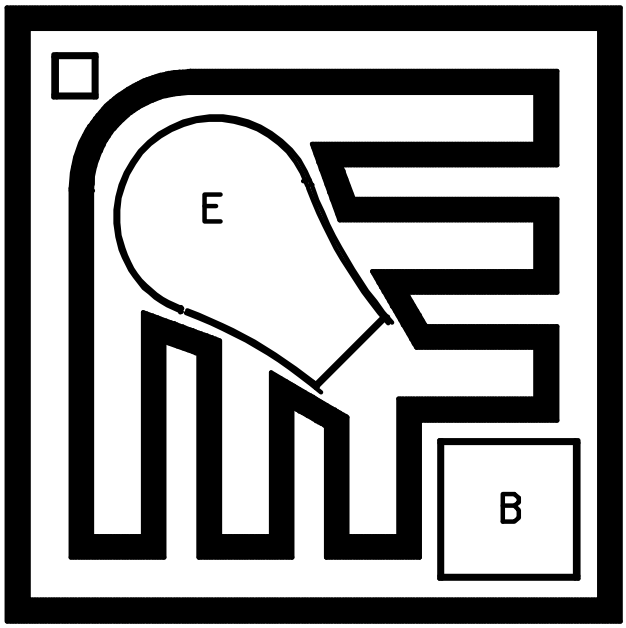
Physical characteristics:
A-version

Die size: 0.021 by 0.021 inches square
Die thickness: 0.010 " 0.001 inches
Base: 0.0044 x 0.0044 inches
Emitter: 0.0041 x 0.0041 inches
Back metal: 3 K to 4.6 KÅ Gold
Top metal: 11K ± 2Å Aluminum
Back side: Collector
Glassivation: Si_3N_4

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 5. JANHC and JANKC (A-version, 2N2907A only) die dimensions.

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Physical characteristics:
B-version

Die size: 0.020 by 0.020 inches square, ± 0.001
Die thickness: 0.0090 ± 0.0015 inches
Base: 0.005 square
Emitter: 0.006 by .004 inches
Back metal: Gold 4.5 KÅ ± 1 KÅ
Top metal: Aluminum 15,000 KÅ, ± 2 KÅ
Back side: Collector
Glassivation: Si_3N_4, 2 kÅ nominal.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified tolerance is ± 0.00100 inch (0.00254 mm).

FIGURE 6. JANHC and JANKC (B-version) die dimensions.

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1.4 Primary electrical characteristics at T_A = +25°C.

h_{FE} at V_{CE} = 10 V dc										
	h_{FE1} I_C = 0.1 mA dc		h_{FE2} I_C = 1.0 mA dc		h_{FE3} I_C = 10 mA dc		h_{FE4} 1/ I_C = 150 mA dc		h_{FE5} 1/ I_C = 500 mA dc	
	2N2906A, L, UA,UB	2N2907A, L, UA,UB	2N2906A L, UA,UB	2N2907A L, UA,UB	2N2906A L, UA,UB	2N2907A L, UA,UB	2N2906A L, UA,UB	2N2907A L, UA,UB	2N2906A L, UA,UB	2N2907A L, UA,UB
Min	40	75	40	100	40	100	40	100	40	50
Max			175	450			120	300		

Types	Limit	$\lvert h_{fe} \rvert$ f = 100 MHz V_{CE} = 20 V dc I_C = 20 mA dc	C_{obo} 100 kHz ≤ f ≤ 1 MHz V_{CB} = 10 V dc I_E = 0	Switching (saturated)	
				t_{on} See figure 7	t_{off} See figure 8
2N2906A, 2N2907A L, UA, UB			pF	ns	ns
	Min	2.5			
	Max		8	45	300

Types	Limits	$V_{CE(sat)1}$ 1/ I_C = 150 mA dc I_B = 15 mA dc	$V_{CE(sat)2}$ 1/ I_C = 500 mA dc I_B = 50 mA dc	$V_{BE(sat)1}$ 1/ I_C = 150 mA dc I_B = 15 mA dc	$V_{BE(sat)2}$ 1/ I_C = 500 mA dc I_B = 50 mA dc
2N2906A, 2N2907A L, UA, UB		V dc	V dc	V dc	V dc
	Min				
	Max	0.4	1.6	1.3	2.6

1/ Pulsed see 4.5.1.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500, and as specified herein.

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and figures 1, 2, 3, 4, 5, and 6 herein. Epoxy die attach may be used when a moisture monitor plan has been submitted and approved by the qualifying activity.

3.4.1 Lead finish. Lead finish shall be solderable as defined in MIL-PRF-19500. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500, except for the UB suffix package. Marking on the UB package shall consist of an abbreviated part number, the date code, and the manufacturers symbol or logo. The prefixes JAN, JANTX, JANTXV, and JANS can be abbreviated as J, JX, JV, and JS respectively. The "2N" prefix and the "AUB" suffix can also be omitted.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3 herein.

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

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4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with MIL-PRF-19500 (table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see appendix E table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3c	Thermal impedance (see 4.3.3)	Thermal impedance (see 4.3.3)
9	I_{CBO2}, h_{FE2}	Not applicable
10	24 hours minimum	24 hours minimum
11	I_{CBO2}; h_{FE2}; ΔI_{CBO2} = 100 percent of initial value or 5 nA dc, whichever is greater. Δh_{FE2} = ±15 percent	I_{CBO2} ,h_{FE4}
12	See 4.3.2, 240 hours minimum	See 4.3.2, 80 hours minimum
13	Subgroups 2 and 3 of table I herein; ΔI_{CBO2} = 100 percent of initial value or 5 nA dc, whichever is greater; Δh_{FE2} = ±15 percent	Subgroup 2 of table I herein; ΔI_{CBO2} = 100 percent of initial value or 5 nA dc, whichever is greater; Δh_{FE4} = ±15 percent

4.3.1 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with MIL-PRF-19500, appendix H. As a minimum, die shall be 100 percent probed to insure the assembled chips will meet the requirements of group A, subgroup 2.

4.3.2 Power burn-in conditions. Power burn-in conditions are as follows: T_A = Room ambient as defined in the general requirements of MIL-STD-750, 4.5.

JANS level (all device types) - - - - - - V_{CB} = 10 - 30 Vdc, P_T = 400 mW (except P_T = 200 mW for UB).
JANTX and JANTXV levels:
2N2906A, L, UA; 2N2907A, L, UA - - - - - V_{CB} = 10 - 30 Vdc, P_T = 400 mW.
2N2906AUB, 2N2907AUB - - - - - V_{CB} = 10 -30 Vdc, P_T = 200 mW.

NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.3.3 Thermal impedance ($Z_{\theta JX}$ measurements). The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, Method 3131.

a. I_M measurement current ..... 5 mA.

b. I_H forward heating current..... 200 mA (min).

c. t_H heating time ..... 25 - 30 ms.

d. t_{md} measurement delay time..... 60 µs max.

e. V_{CE} collector-emitter voltage..... 10 V dc minimum

The maximum limit for $Z_{\theta JX}$ under these test conditions are $Z_{\theta JX}$ (max) = 72°C/W.

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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. If alternate screening is being performed in accordance with E.5.3.1d of MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) of MIL-PRF-19500 and 4.4.2.1. Electrical measurements (end-points) and delta requirements shall be in accordance with group A, subgroup 2 and 4.5.3 herein. See 4.4.2.2 for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) and delta requirements for JAN, JANTX, and JANTXV shall be after each step in 4.4.2.2 and shall be in accordance with group A, subgroup 2 and 4.5.3 herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B4	1037	V_{CB} = 10 - 30 V dc; T_J = 150°C; t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No heat sink or forced-air cooling on devices shall be permitted.
B5	1027	V_{CB} = 10 - 30 V dc; T_A = 100°C for 96 hours, or T_A = +125°C ± 25°C for 96 hours with P_T adjusted according to the chosen T_A to give an average T_J = +275°C.

4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV). 1/

Step	Method	Condition
1	1039	Steady-state life: Test condition B, 340 hours, V_{CB} = 10 - 30 V dc, T_J = +150°C minimum. No heat sink or forced-air cooling on devices shall be permitted. n = 45, C = 0.
2	1039	The steady-state life test of step 1 shall be extended to 1,000 hours for each die design. Samples shall be selected from a wafer lot every twelve months of wafer production. Group B step 2 shall not be required more than once for any single wafer lot. n = 45, C = 0.
3	1032	High- temperature life (non operating), T_A = +200°C, t = 340 hours, n = 22, C = 0.

4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. For JANS, samples shall be selected from each inspection lot. See MIL-PRF-19500.

b. Must be chosen from an inspection lot that has been submitted to and passed group A, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B4 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and in 4.4.3.1 (JANS).and 4.4.3.2 (JAN, JANTX, and JANTXV) herein for group C testing. Electrical measurements (end points) and delta requirements shall be in accordance with group A, subgroup 2 and 4.5.3 herein.

1/ Separate samples may be used for each step. In the event of a group B failure, the manufacturer may pull a new sample at double size from either the failed assembly lot or from another assembly lot from the same wafer lot. If the new “assembly lot” option is exercised, the failed assembly lot shall be scrapped.

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4.4.3.1 Group C inspection, table VII (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E; not applicable for UA and UB devices.
C6	1026	1,000 hours at V_{CB} = 10 - 30 V dc; T_J = +150°C minimum. No heat sink or forced-air cooling on device shall be permitted.

4.4.3.2 Group C inspection, table VII (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E. Test condition E; not applicable for UA and UB devices.
C6		Not applicable.

4.4.3.3 Group C sample selection. Samples for steps in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes group A tests for conformance inspection. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.4 Group E inspection. Group E inspection shall be performed for qualification or requalification only. The tests specified in table II herein must be performed to maintain qualification.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Input capacitance. This test shall be conducted in accordance with method 3240 of MIL-STD-750, except the output capacitor shall be omitted.

4.5.3 Delta requirements. Delta requirements shall be as specified below:

Step	Inspection	MIL-STD-750		Symbol	Limit	Unit
		Method	Conditions			
1	Collector-base cutoff current	3036	Bias condition D, V_{CB} = 50 V dc	ΔI_{CB02} 1/	100 percent of initial value or 10 nA dc, whichever is greater.	
2	Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1.0 mA dc; pulsed see 4.5.1	Δh_{FE2} 1/	±25 percent change from initial reading.	

1/ Devices which exceed the group A limits for this test shall not be accepted.

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TABLE I. Group A inspection

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical inspection 3/	2071	n = 45 devices, c = 0				
Solderability 3/ 5/	2026	n = 15 leads, c = 0				
Resistance to solvents 3/ 4/ 5/	1022	n = 15 devices, c = 0				
Temp cycling 3/ 5/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
Heremetic seal 5/ Fine leak Gross leak	1071	n = 22 devices, c = 0				
Electrical measurements 5/		Group A, subgroup 2				
Bond strength 3/ 5/	2037	Precondition T_A = + 250°C at t = 24 hours or T_A = + 300°C at t = 2 hours n = 11 wires, c = 0				
Subgroup 2						
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 60 V dc	I_{CBO1}		10	uA dc
Cutoff current, emitter to base	3061	Bias condition D; V_{EB} = 5 V dc	I_{EBO1}		10	uA dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	60		V dc
Collector to emitter cutoff current	3041	Bias condition D; V_{CE} = 50 V dc	I_{CES}		50	nA dc
Collector to base cutoff Current	3036	Bias condition D; V_{CE} = 50 V dc	I_{CBO2}		10	nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 4 V dc	I_{EBO2}		50	nA dc
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 0.1 mA dc	h_{FE1}			
2N2906A, L, UA, UB 2N2907A, L, UA, UB				40 75		

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued	3076	V_{CE} = 10 V dc; I_C = 1.0 mA dc	h_{FE2}			
Forward-current transfer ratio						
2N2906A, L, UA, UB 2N2907A, L, UA, UB				40 100	175 450	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc	h_{FE3}			
2N2906A, L, UA, UB 2N2907A, L, UA, UB				40 100		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc; pulsed (see 4.5.1)	h_{FE4}			
2N2906A, L, UA, UB 2N2907A, L, UA, UB				40 100	120 300	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 500 mA dc; pulsed (see 4.5.1)	h_{FE5}			
2N2906A, L, UA, UB 2N2907A, L, UA, UB				40 50		
Collector-emitter saturation voltage	3071	I_C = 150 mA dc; I_B = 15 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.4	V dc
Collector-emitter saturation voltage	3071	I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$		1.6	V dc
Base-emitter saturation voltage	3066	Test condition A; I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)1}$	0.6	1.3	V dc
Base-emitter saturation voltage	3066	Test condition A; I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)2}$		2.6	
Subgroup 3						
High temperature operation		T_A = +150°C				
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 50 V dc	I_{CBO3}		10	µA dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 3 continued						
Low temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc	h_{FE6}			
2N2906A, L, UA, UB				20		
2N2907A, L, UA, UB				50		
Subgroup 4						
Small-signal short- circuit forward current transfer ratio	3206	V_{CE} = 10 V dc; I_C = 1 mA dc; f = 1 kHz	h_{fe}			
2N2906A, L, UA, UB				40		
2N2907A, L, UA, UB				100		
Magnitude of small- signal short- circuit forward current transfer ratio	3306	V_{CE} = 20 V dc; I_C = 20 mA dc; f = 100 MHz	$\|h_{fe}\|$	2.0		
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		8	pF
Input capacitance (output open- circuited)	3240	V_{EB} = 2.0 V dc; I_C = 0; 100 kHz ≤ f ≤ 1 MHz	C_{ibo}		30	pF
Saturated turn-on time		(See figure 7)	t_{on}		45	ns
Saturated turn-off time		(See figure 8)	t_{off}		300	ns
Subgroups 5 and 6						
Not applicable						
Subgroup 7 4/						
Decap internal visual (design verification)	2075	n = 1 device, c = 0				

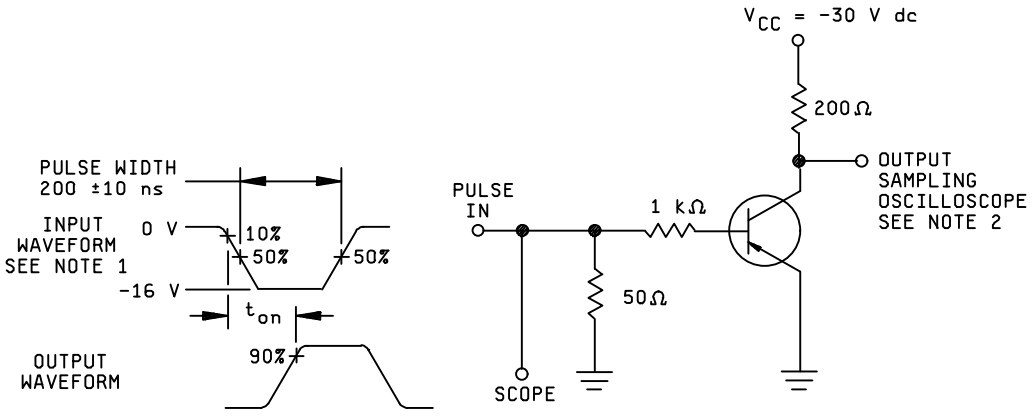
1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed subgroup A1, double the sample size of the failed test or sequence of tests. A failure in group A, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for laser marked devices.
5/ Not required for JANS devices.

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TABLE II. Group E inspection (all quality levels) - For qualification only

Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			12 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See group A, subgroup 2 and 4.5.3 herein.	
Subgroup 2			45 devices c = 0
Intermittent life	1037	Intermittent operation life: V_{CB} = 10 V dc , 6,000 cycles, $\Delta T_J \geq +100°C$; forced air cooling allowed on cooling cycle only.	
Electrical measurements		See group A, subgroup 2 and 4.5.3 herein.	
Subgroup 3			
Not applicable			
Subgroup 4			
Not applicable			
Subgroup 5			
Not applicable			

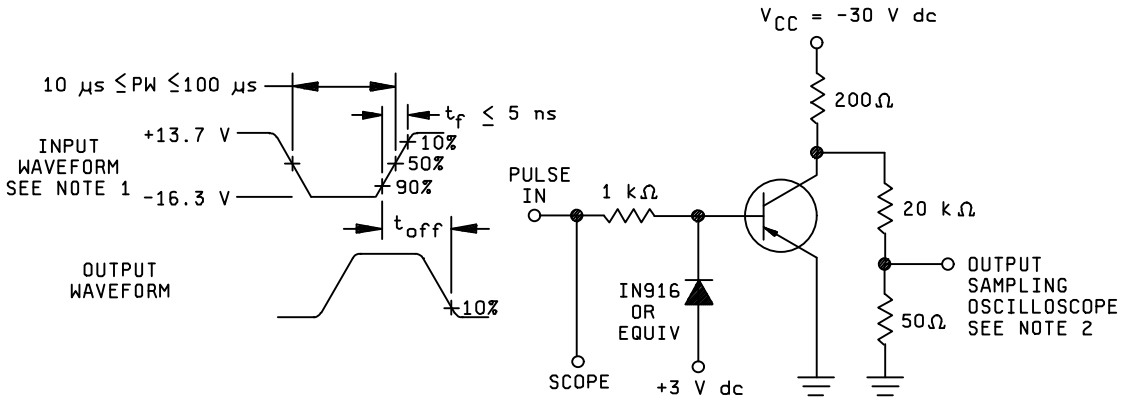
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NOTES:

1. The rise time (t_r) of the applied pulse shall be ≤ 2.0 ns, duty cycle ≤ 2 percent and the generator source impedance shall be 50 ohms.
2. Sampling oscilloscope: Z_{in} ≥ 100 K ohms, C_{in} ≤ 12 pF, rise time ≤ 5 ns.

FIGURE 7. Saturated turn-on switching time test circuit.



NOTES:

1. The rise time (t_r) of the applied pulse shall be ≤ 2.0 ns, duty cycle ≤ 2 percent and the generator source impedance shall be 50 ohms.
2. Sampling oscilloscope: Z_{in} ≥ 100 K ohms, C_{in} ≤ 12 pF, rise time ≤ 5 ns.

FIGURE 8. Saturated turn-off switching time test circuit.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-PRF-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. See MIL- PRF-19500.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, DSCC-VQE, Columbus, OH 43216.

6.4 Supersession information. Devices covered by this specification supersede the manufacturers' and users' Part or Identifying Number (PIN). The term Part or Identifying Number (PIN) is equivalent to the term part number which was previously used in this specification. This information in no way implies that manufacturers' PIN's are suitable as a substitute for the military PIN.

6.5 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCA2N2907A) will be identified on the QPL.

Die ordering information 1/		
PIN	Manufacturer	
	32694	43611
2N2906A 2N2907A	JANHCA2N2907A	JANHCB2N2906A JANHCB2N2907A

1/ For JANKC level, replace JANHC with JANKC.

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 13, 19, 85, 99

Preparing activity:
DLA - CC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/291G	2. DOCUMENT DATE (YYMMDD) 980831

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, SWITCHING, TYPES 2N2906A, 2N2906AL, 2N2907A, 2N2907AL, 2N2906AUA 2N2907AUA, 2N2906AUB, AND 2N2907AUB JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) Commercial DSN FAX EMAIL	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. Point of contact: Alan Barone,	b. TELEPHONE Commercial 614-692-0510 DSN 850-0510 FAX 614-692-6939 EMAIL alan_barone@dscc.dla.mil
c. ADDRESS : Defense Supply Center Columbus, ATTN: DSCC-VQE, 3990 East Broad Street, Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

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